# Parallel and Distributed Computing

Dr. Imran Ashraf

imran.ashraf@hitecuni.edu.pk

# FLOPS

- Floating point operations per second
- Measure of theoretical peak performance

$$\mathbf{FLOPS} = \mathbf{sockets} \times \frac{\mathbf{cores}}{\mathbf{socket}} \times \frac{\mathbf{cycles}}{\mathbf{second}} \times \frac{\mathbf{FLOPs}}{\mathbf{cycle}}$$

# FLOPS

- Servers are the only computers that sometimes have more than one socket; for most home computers (desktop or laptop), “sockets” will be 1.
- Cores per socket depends on your CPU. It could be 2 (dual-core), 3, 4 (quad-core), 6 (hexa-core), or 8. There are some prototype CPUs with as many as 80 cores.
- “Clock cycles per second” refers to the speed of your CPU. Most modern CPUs are rated in gigahertz. So 2 GHz would be 2,000,000,000 clock cycles per second.
- The number of FLOPs per cycle also depends on the CPU. One of the fastest (home computer) CPUs is the Intel Core i7–970, capable of 4 double-precision or 8 single-precision floating-point operations per cycle.

# Test

- Intel Core i7–970 has 6 cores. If it is running at 3.46 GHz and can perform 8 floating point operations per second, calculate the theoretical compute power of this machine.

# Example

- Intel Core i7–970 has 6 cores. If it is running at 3.46 GHz, the formula would be:
- 1 (socket) * 6 (cores) * 3,460,000,000 (cycles per second) * 8 (single-precision FLOPs per second) = 166,080,000,000 single-precision FLOPs per second or 83,040,000,000 double-precision FLOPs per second.
- 109 GFLOPS.

# Speedup calculations

- Ratio
- %age

- Upper bounds on speedup

# Amdahl's Law

- Theoretical speedup calculations

$$S_{\text{latency}}(s) = \frac{1}{(1-p) + \frac{p}{s}}$$

# Example 1

- If 30% of the execution time may be the subject of a speedup, *p* will be 0.3; if the improvement makes the affected part twice as fast, *s* will be 2. Amdahl's law states that the overall speedup of applying the improvement will be?

# Example 2

- Assume that we are given a serial task which is split into four consecutive parts, whose percentages of execution time are $p1 = 0.11$, $p2 = 0.18$, $p3 = 0.23$, and $p4 = 0.48$ respectively. Then we are told that the 1st part is not sped up, so $s1 = 1$, while the 2nd part is sped up 5 times, so $s2 = 5$, the 3rd part is sped up 20 times, so $s3 = 20$, and the 4th part is sped up 1.6 times, so $s4 = 1.6$. By using Amdahl's law, the overall speedup is?

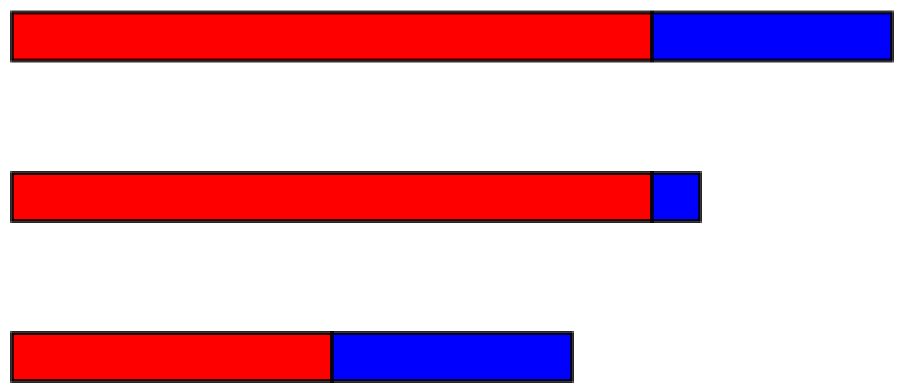
# Amdahl's Law

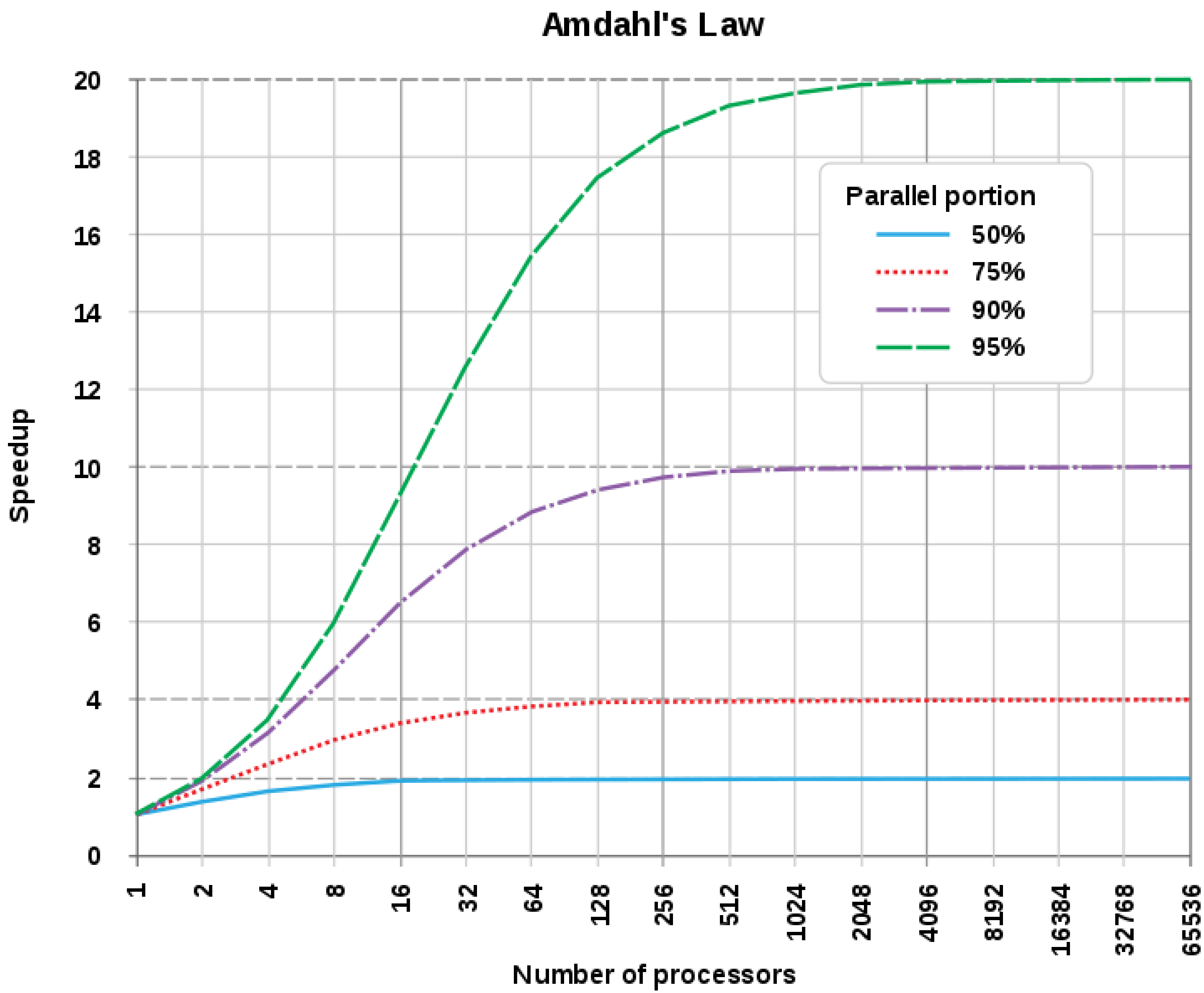
Two independent parts A B

Original process

Make B 5x faster

Make A 2x faster




Amdahl's Law
Speedup
20
18
16
14
12
10
8
6
4
2
0
1
2
4
8
16
32
64
128
256
512
1024
2048
4096
8192
16384
32768
65536
Number of processors
Parallel portion
50%
75%
90%
95%

# Useful Reading

- https://lwn.net/Articles/250967/
- https://people.cs.clemson.edu/~dhouse/courses/405/papers/optimize.pdf

# Thanks